

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)	KE + gravitational PE = 0 & context explained briefly (also accept KE = gravitational PE and context) (1) Minimum context = conservation of energy or mention of galaxy or spherical universe or uniform universe or cosmological principle or explaining why $r = D$ or explaining what the mass of the sphere is (accept whole universe or part and mass inside) Reference to universe nor expanding now loses 1 st mark $v = H_0 D$ substituted for v (1) mass = density $\times$ volume of sphere (1) Algebra set out clearly (1)	4			4	3	
		(ii)	Correct substitution into formula (gives 8.66×10^{-27}) (1) Dividing by 1.66×10^{-27} (or 5.2 or 5.22 seen) OR $5u = 8.3 \times 10^{-27}$ (1)	1	1		2	1	
	(b)	(i)	Halfway between or correct substitutions into C of M equation leading to 1×10^{11} or simply 1×10^{11} [m]		1		1		
		(ii)	Substitution into period equation (1) 9.73×10^6 s (allow 1 mark for 13.8×10^6 OR 4.87×10^6 OR 3.44×10^6) (1)	1	1		2	2	

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		(iii)	Use of $v = \frac{2\pi r}{T}$ ecf or equivalent with $v = \omega r$ and $\omega = \frac{2\pi}{T}$ (1) (expect 65 000 m s ⁻¹) Use of Doppler equation (1) Correct answer = 0.0934 nm (accept 1sf) (1) Note: ecf on previous values gives 0.187 nm, 0.264 nm and 0.066 nm (for full marks unless another clear mistake seen) Allow 2 marks max if d used instead of r (gives 0.187 nm) Allow 1 mark max if $\frac{1}{2}mv^2 = \frac{GMm}{r}$ (also gives 0.187 nm)	1 1	 1		3	3	
	(c)		Centre of mass is nearer more massive star (1) Which will give a larger field OR but must be nearer smaller star for fields to match (1) Accept equivalent reverse argument		2		2	2	
	(d)	(i)	Force [due to both stars] are always equal [and opposite] Accept this is the point where the field is zero Accept masses of equal mass and distance either side		1		1		
		(ii)	Force due to black hole is 4×greater or 4.17×10^{30} [N] (1) Also force from other star (gives factor 5) or 1.04×10^{30} [N] (1)		2		2	1	

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		(iii)	Increasing v increases redshift OR linking orbital speed to redshift (can be implied by good answers) (1) Correct realisation that factor is $\sqrt{5}$ or that v^2 proportional to force (this implies the 1 st mark) (1) Joseff is wrong (based on some valid physics) (1) e.g. red shift increases with orbital speed but they aren't proportional (force and speed) so Jo is wrong (this gets 2 marks – 1 st and 3 rd) Also accept for 2 marks: Red shift increases with orbital speed and orbital speed is proportional to force, so Jo is right Also accept calculations: Expected $v = 144\,000\text{ m s}^{-1}$ (1) Red shift = 0.21 n[m] (1) Joseff is wrong (1)			3	3	2	
	(e)		[More] experiments involving Higgs [decay] (1) Results [need to be] in agreement with theory (1)			2	2		
			Question 5 total	8	9	5	22	12	0